BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MIRPURKHAS

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section-A (MCQ's)

Note: Q.1: Choose the correct answer for the the given option:

Qi. $(ax^3+ay^3) / (a^2(x+y)) =$ _____

(a) $ax^2+ ay^2$
(b) $x^2 + y^2$
(c) $a(x^2 - xy + y^2)$
(d) $a (x^2 +xy + y^2)$

Qii. _____ is th e equidistainct for all these sides of a triangle.

(a) incentre
(b) Circumcentre
(c) Orthocentre
(d) Cntroid

Qiii. Base in Natural logearithm is _____

(a) 10
(b) e
(c) π
(d) 5

Qiv. How many maximum rootss of a Quadratic equation are:

(a) 2
(b) 3
(c) 5
(d) 4

Qv. PQR - m∠P = 65o and m∠Q = 57° then _____ will be teh shortest side.

(a) $P\bar{Q}$
(b) $Q\bar{R}$
(c) $R\bar{P}$
(d) None of these

Qvi. Factors of $a^2 + 2ab + b^2 - c^2$ is:

(a) (a - b - c)(a - b - c)
(b) (a + b + c)(a - b - c)
(c) (a + b + c)(a + b - c)
(d) (a + b + c)(a - b - c)

Qvii. The line x = - 5 is paralle to :

(a) x - axis
(b) y - axis
(c) both axis
(d) None of these

Qviii. $(a + b)(a - b)(a^2 - b^2) =$ _____.

(a) $(a^2 - b^2)^2$
(b) $a^4 - b^4$
(c) $(a^2 + b^2)$
(d) $(a^4 + b^4)$

Qix. Solution se of $\sqrt{y} = 10$ is :

(a) {100}
(b) {10}
(c) {-10}
(d) {-10,10}

Qx. The bisector of the angles of a traingle are:

(a) Concrurrent
(b) Collinear
(c) Unequal
(d) Do not inersect

Qxi. Standard form of Quadratic equation is:

(a) $ax^2 + bx + c$ 0, $a \neq 0$
(b) $ax^2 + c = 0$, $a \neq 0$
(c) $ax^2 + bx = 0$ $a \neq 0$
(d) $ax^3 + bx^2 + c = 0$, $a \neq 0$

Qxii. $\sqrt{y}+4+2=8$ is a :

(a) linear equation
(b) Radical equation
(c) cubic equation
(d) Quadratic equation

Qxiii. A triangle havign all the three sides congruent is called _____ triangle.

(a) Scalene
(b) Right -angled
(c) equilateral
(d) Isosceles

Qxiv. HCF of $x^3 - 83$ and $x^2 - 4xy + 4y^2$ is:

(a) x - 4y
(b) $x^2 + 2xy + y^2$
(c) x + 2y
(d) x - 2

Qxv. QUQ' = _____

(a) R
(b) E
(c) N
(d) W

Qxvi. Scientifc notation of 0.00789 is _____.

(a) 7.89×10^3
(b) 7.89 x 10
(c) 0.789x102
(d) 78.9×10^4

Qxvii. The value of 't' for which $49x^2 - 42xy + 1$ will be a prefect square:

(a) 9
(b) $9y^2$
(c) $3y^2$
(d) 3

Qxviii. $(5 + \sqrt{5})(5 - \sqrt{15})$ is equal to:

(a) 10
(b) 15
(c) 25
(d) 30

Qxix. E^4 is closed under:

(a) Division
(b) Multiplication
(c) Substraction
(d) All of these

Qxx. (5i) (-2i) = _____

(a) -10
(b) 10
(c) -10i
(d) 10i

Qxxi. log a/bcd = _____.

(a) loga -log b + logc + logd
(b) loga - logb + logc - logd
(c) loga - logb - logc - logd
(d) loga - log - logc + logd

Qxxii. The degree of the $3^3y + 5y^4 - 10$ is

(a) 4
(b) 5
(c) 6
(d) 10

Qxxiii. The _____ of a triangle cut each other in the ratio 2:1 :

(a) Altitude
(b) Median
(c) angle bisector
(d) Right bisector

Qxxiv. If the sum of measures o∠A and ∠C of a parallelogram ABCD is 140o then ∠D will be of _____.

(a) 70°
(b) 10°
(c) 140°
(d) 220°

Qxxv. $((x_1+x_2)/2, (y_1+y_2)/2)$ is called:

(a) Mid point formula
(b) Distance formula
(c) Division formula
(d) Ratio formola

Qxxvi. _____ are the length of sides ofa right angle triangle .

(a) 3, 4, and 5
(b) 1, 3, and 5
(c) 2, 4, and 6
(d) None of these

Qxxvii. A point equidistant fom the end points of a line segment is on the _____ of it.

(a) Perpendicular
(b) Right bisector
(c) Centre
(d) Bisector

Qxviii. The point (-3, -4) is located in:

(a) 1st Quadrant
(b) 2nd Quadrant
(c) 3rd Quadrant
(d) 4th Quadrant

Qxxix. If $\log_{10} x = 4$, then x = _____

(a) 500
(b) 100
(c) 1000
(d) 10000

Qxxx. In siplified form $1/(x+y) + y/(x^2+y^2)$ = _____

(a) $(y+1)/(x^2+y^2)$
(b) $x/(x^2+y^2)$
(c) y/x^2
(d) $1/(x^2 - y^2)$

Section-B

> Note: Attempt any NINE of the following questions:

Q2. Simplify:

(i) $2^4\ 5^3 / 10^2$

(ii) $\{3(-4)\}^3$

Q3. If Log 3 = 0.4717, Log 2 = 0.3010, Log 5 = 0.6990 then find the value of Log [illegible in source scan]

Q4. If ab = 5 and ab=7, find the value of a^3b^3. Find the factors of the following

Q5.

(i) $x^2 - 1 + 1x^2/4x^2$

(ii) $(2x + x)2 - (2x -x)^2$

Q6. Find the LCM of $x^2 +11x + 28$ and $x^2 + x - 12$

Q7. Simplify

(ii) $\sqrt{x} - 8 = 1$

(ii) $|5x - 12| = 7$

Q8. Solve $9x^2 = 12x - 49$ by quadratic formula

Q9. solve $\sqrt{(x+2)} + \sqrt{(x+7)} = \sqrt{(6x+13)}$

Q10. Find the value of x and y when $x + yt = -5 + 5i$

Q11. $x = 3 - 2\sqrt{2}$ find $x^2 + 1/x^2$

Q12. If the polynomial $4x^3 - 7x^2 + 6x - 3k$ is exactly divisible by $(x - 2)$ find the value of K

Q13. Find the square root by division method of $2x^2\ 3x^2 + 2x + 1$

Q14. Find the number of digits in 45.

SECTION - C

> Note: ATT9AP: ANT I-015E1010F THE fOLLOwsit3 QUESTIONS.

Q15. If two angles of a triangle are congruent then opposite sides will also be congruent

Q16. Define with Diagram

(i) Right Bisector of fine segment

(ii) Equilateral Triangle

(iii) Congruent Triangles

Q17. If opposite sides of a quadrilateral are congruent and parallel then quadrilateral will be parallelogram.

Q18. Draw $\triangle ABC$. if $m\angle A = 65°$ m.AC = mAB = 5 lcm

Q19. Find am central point of a circle when the end points of diameter are A(1,5) and B(3-4).